SUPPLEMENTARY MATERIAL - QUANTIFYING DRUG-INDUCED BONE MARROW TOXICITY USING A NOVEL HAEMATOPOIESIS SYSTEMS PHARMACOLOGY MODEL

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SUPPLEMENTARY TEXT

S1. Animals and study design

Male RccHan:WIST rats were supplied by Harlan UK. Animals were approximately 12 weeks old with a minimum 320g of body weight at the start of dosing. Up to 10 rats per treatment group were randomised (by cages) based on results of haematology obtained 7 days prior to the study start.

Animals were dosed intravenously (iv) via tail vein by 1-minute injection, with either vehicle or carboplatin (Tocris), formulated as a solution in 0.9% w/v physiological saline, dose volume 5ml/kg. In multiple cycle studies, carboplatin / vehicle was dosed every 14 days (Table S1). Blood samples were taken at selected timepoints (Table S1) for either haematology (supplementary text S2) or free platinum concentration (supplementary text S4) from the tail vein (0.4 mL into EDTA) or from the vena cava at necropsy, when sampling was within 1 day from dosing. At termination, rats were euthanised by administration of halothane. Femurs were removed at the necropsy to perform flow cytometry analysis of bone marrow cells (supplementary text S3).

S2. Haematology analysis

Haematology analysis was performed using the Siemens Advia 2120i haematology analyser, on days reported in Table S1. The following haematology parameters were measured: erythrocytes, haemoglobin, haematocrit, mean red cell haemoglobin, mean red cell haemoglobin concentration, mean red cell volume, red cell distribution width, reticulocytes, platelets, leucocytes, neutrophils, lymphocytes, monocytes, basophils, eosinophils, and large unstained cells.

S3. Flow cytometry analysis of bone marrow cells

Bone marrow cells were immediately flushed out of trimmed rat femurs with 3 ml PBS containing 50% fetal calf serum. Cell suspension was syringed and filtered through 100 µm strainer, collected by centrifugation (300 g/ 7 min/ 4ºC) and washed once in HBSS containing 2% FCS and 10 mM HEPES (staining buffer). Total cell count of isolated cells was determined by automated cell counter (Countess, Invitrogen). Cell concentration was adjusted to $1x10^7$ cells/ ml in staining buffer and processed for antibody staining. (1) CD90.1 and Lineages cocktail: 1 ml of cell suspension was resuspended in 100 µl staining buffer containing anti-rat CD90.1-APC (dilution 1:100), CD6-FITC (1:100), CD3-FITC (1:100), CD11b-FITC (1:200), Granulocytes-FITC (1:200) from BD Pharmingen and CD45RC-FITC (1:100) purchased from Serotec. (2) CD71 and CD45 cocktail: 100 µl cell suspension was resuspended in 100 µl staining buffer containing anti-rat CD71-FITC (dilution 1:10) and CD45-PE (1:10) from Serotec. Cells were incubated with antibodies for 30 minutes at room temperature and washed twice in staining buffer. After final centrifugation, 1 ml of staining buffer was added to CD90.1-Lineages stained cells. Cell pellet stained with CD71-CD45 antibodies was resuspended in 200 µl staining buffer. This was followed by addition of 10 µl of LDS-751 cell-permeant nuclear stain (Life Technologies). Cells were incubated in dark for 30 minutes prior to flow cytometry analysis. Data (at least 10,000 events) were acquired on FACS AriaII (BD). Analysis was performed in FlowJo software. CD90+/Lineage-, CD45+, and CD71+ cells were used to quantify multi-potent progenitor (MPP), common myeloid progenitor (CMP) and megakaryocyte-erythrocyte progenitor (MEP) populations, respectively.

S4. Plasma and bone marrow free platinum (carboplatin active species) bioanalysis in rat

Each plasma sample (100 µl) was added into amicon ultra-0.5 3kDa 0.5ml centrifuge tubes alongside an 11-point standard calibration curve (1-10000 nM) prepared in DMSO and spiked into blank plasma. Samples were centrifuged at 13000rpm for 10mins and the ultrafiltrate was transferred to a processing plate. 50ul Paladium acetate (10uM in water) was spiked into the platinum fortified plasma ultrafiltrate. The addition of 50 µL 5% diethyldithiocarbamate (DDTC) in 0.1N NaOH into all samples was followed by a 30min incubation at 45^{o}C. Liquid-Liquid extraction is carried out with the addition of MTBE, shaking for 30mins and extraction of the organic layer. The organic layer is transferred to a clean plate and evaporated off under nitrogen. Samples were reconstituted in Acetonitrile / Water (50:50) and analysed by UPLC MSMS.

All Femurs were washed through with 2ml PBS buffer & collected into tubes. These were separated by centrifugation & removal of the supernatant to a separate tube. The number of cells were counted for each of the bone marrow pellets. Each supernatant sample (50ul) was removed into a DWP containing 200ul of ice cold Acetonitrile containing IS and vortexed for 30 seconds. Samples were centrifuged at 3000rpm for 15mins at 4^{o}C, 50 µl supernatant removed derivatives with DDTC as with the plasma and analysed by UPLC-MSMS. The cell pellet was re-suspended in 50 µL PBS and vortexed for 30 secs, it is then treated as supernatant prior to analysis. An 11-point standard calibration curve (1-10000 nM) prepared in DMSO and spiked into blank supernatant/ resuspended cells taken from untreated animals.

S5. Free platinum (carboplatin active species) pharmacokinetic model in rat

Based on in-house data (carboplatin 30 mg/kg i.v. dose, Table S1), and literature data from Siddik *et al*. [1] (carboplatin 20 mg/kg i.v. dose), rat plasma free platinum pharmacokinetics was described with the following 2 compartment PK model:

$$\frac{dCen}{dt} = -\frac{CL}{V_{Cen}}Cen - \frac{Q_C}{V_{Cen}}Cen + \frac{Q_C}{V_{Per}}Per,$$

$$\frac{dPer}{dt} = \frac{Q_C}{V_{Cen}}Cen - \frac{Q_C}{V_{Per}}Per,$$

$$Cp = \frac{Cen}{V_{Cen}}.$$

Equation S1

Where *Cen* and *Per* describe the amount of free platinum in the central and peripheral compartment, respectively, while C_p describes free platinum concentration in the plasma. V_{Cen} and V_{Per} are volumes of distribution for the central and peripheral compartments, $k_{el} = CL / V_{Cen}$ is a first order elimination rate from the central compartment, and $k_{12} = Q_C / V_{Cen}$, $k_{21} = Q_C / V_{Per}$ are first-order intercompartmental exchange rates.

S6. Homeostasis conditions

In the absence of perturbations - such as those induced by carboplatin, the system of ordinary differential equations (ODEs) described in Equation 1 – Equation 9 (main text), is at steady state. In other words, cell populations remain constant at their physiological levels:

$$
\begin{aligned}
&MPP = MPP_0, CMP = CMP_0, MEP = MEP_0,\\
&T1_{Neut} = T1_{Neut_0}, T2_{Neut} = T2_{Neut_0}, T3_{Neut} = T3_{Neut_0},\\
&T1_{Mono} = T1_{Mono_0}, T2_{Mono} = T2_{Mono_0}, T3_{Mono} = T3_{Mono_0},\\
&T1_{Plt} = T1_{Plt_0}, T2_{Plt} = T2_{Plt_0}, T3_{Plt} = T3_{Plt_0},\\
&T1_{Ret} = T1_{Ret_0}, T2_{Ret} = T2_{Ret_0}, T3_{Ret} = T3_{Ret_0},\\
&Neut = Neut_0, Mono = Mono_0,\\
&Plt = Plt_0, Ret = Ret_0, RBC = RBC_0.
\end{aligned}
$$

Equation S2

In order to define conditions necessary to guarantee this equilibrium, we imposed the right-hand side of in Equation 1, 4, 5, 7, 9 (main text) equal to zero, and we solved algebraically these set of equations.

This resulted in the following set of steady state equations:

A. Transit compartment baseline values:

$$
\begin{aligned}
&T1_{Neut_0} = T2_{Neut_0} = T3_{Neut_0} = T_{Neut_0} = \frac{k_{circ_{Neut}}}{a_{Neut}} Neut_0,\\
&T1_{Mono_0} = T2_{Mono} = T3_{Mono_0} = T_{Mono_0} = \frac{k_{circ_{Mono}}}{a_{Mono}} Mono_0,\\
&T2_{Ret_0} = T3_{Ret} = T_{Ret_0} = \frac{k_{circ_{Ret}}}{a_{Ret}} Ret_0,\\
&T1_{Ret_0} = T1_{Ret_0} = \frac{k_{circ_{Ret}}}{a_{Ret}} \frac{Ret_0}{\lambda_1},\\
&T2_{Plt_0} = T3_{Plt_0} = T_{Plt_0} = \frac{k_{circ_{Plt}}}{a_{Plt}} Plt_0,\\
&T1_{Plt_0} = T1_{Plt_0} = \frac{k_{circ_{Plt}}}{a_{Plt}} \frac{Plt_0}{\lambda_2}.
\end{aligned}
$$

Equation S3

Where λ_1, and λ_2 represent the ratio between $T1_0$ and $T2_0$ for reticulocytes and platelets respectively (Equation S7).

B. Proliferation and maturation rates:

$$k_{circ_{Ret}} = k_{circ_{RBC}} \frac{RBC_0}{Ret_0},$$

$$k_{prol_{CMP}} = \frac{1}{\lambda_3}\left(k_{circ_{Neut}} \frac{Neut_0}{CMP_0} + k_{circ_{Mono}} \frac{Mono_0}{CMP_0} \right),$$

$$k_{prol_{MEP}} = \frac{1}{\lambda_4}\left(\frac{a_{Ret}^2 T1^2{}_{Ret_0}}{k_{circ_{Ret}} Ret_0 MEP_0} + \frac{a_{Plt}^2 T1^2{}_{Plt_0}}{k_{circ_{Plt}} Plt_0 MEP_0} \right),$$

$$k_{tr_{Neut}} = k_{circ_{Neut}} \frac{Neut_0}{CMP_0},$$

$$k_{tr_{Mono}} = k_{circ_{Mono}} \frac{Mono_0}{CMP_0},$$

$$k_{tr_{Ret}} = a_{Ret} \frac{T1_{Ret_0}}{MEP_0} \frac{T1_{Ret_0}}{T_{Ret_0}},$$

$$k_{tr_{Plt}} = a_{Plt} \frac{T1_{Plt_0}}{MEP_0} \frac{T1_{Plt_0}}{T_{Plt_0}},$$

$$k_{tr\,CMP} = (k_{tr_{Neut}} + k_{tr_{Mono}} - k_{prol_{CMP}}) \frac{CMP_0}{MPP_0},$$

$$k_{tr_{MEP}} = (k_{tr_{Ret}} + k_{tr_{Plt}} - k_{prol_{MEP}}) \frac{MEP_0}{MPP_0},$$

$$k_{prol_{MPP}} = \frac{k_{tr_{CMP}} + k_{tr_{MEP}}}{\lambda_5},$$

$$k_{prol_{Ret}} = a_{Ret}\left(1 - \frac{T1_{Ret_0}}{T_{Ret_0}} \right),$$

$$k_{prol_{Plt}} = a_{Plt}\left(1 - \frac{T1_{Plt_0}}{T_{Plt_0}} \right),$$

$$k_{stem} = (k_{tr_{CMP}} + k_{tr_{MEP}} - k_{prol_{MPP}}) MPP_0.$$

Equation S4

Where λ_3, λ_4, and λ_5 denote the ratios among maturation and proliferation rates for CMP, MEP and MPP, respectively (Equation S7).

S7. Reparameterization to set physiological boundaries for parameter estimation

From Equation S4 it follows that, to guarantee positivity for $k_{prol_{Ret}}$ and $k_{prol_{Plt}}$, we needed $T1_{Ret_0}$, T_{Ret_0}, $T1_{Plt_0}$, and T_{Plt_0} to satisfy the following conditions:

$$1\text{-}\frac{T1_{Ret_0}}{T_{Ret_0}} > 0 \Rightarrow \frac{T1_{Ret_0}\,a_{Ret}}{Ret_0 k_{circ_{Ret}}} < 1 \Rightarrow T1_{Ret_0} < \frac{Ret_0 k_{circ_{Ret}}}{a_{Ret}} \Rightarrow 1 < \frac{Ret_0 k_{circ_{Ret}}}{T1_{Ret_0}\,a_{Ret}},$$
$$1\text{-}\frac{T1_{Plt_0}}{T_{Plt_0}} > 0 \Rightarrow \frac{T1_{Plt_0}\,a_{Plt}}{Plt_0 k_{circ_{Plt}}} < 1 \Rightarrow T1_{Plt_0} < \frac{Plt_0 k_{circ_{Plt}}}{a_{Plt}} \Rightarrow 1 < \frac{Plt_0 k_{circ_{Plt}}}{T1_{Plt_0}\,a_{Plt}}.$$

Equation S5

Similarly, to guarantee the positivity of $k_{tr_{CMP}}$, $k_{trl_{MEP}}$, and k_{stem}, in Equation S4, we needed $k_{prol_{CMP}}$, $k_{prol_{MEP}}$, and $k_{prol_{MPP}}$, to satisfy the following conditions:

$$k_{tr_{Neut}} + k_{tr_{Mono}} - k_{prol_{CMP}} > 0 \Rightarrow \frac{k_{tr_{Neut}} + k_{tr_{Mono}}}{k_{prol_{CMP}}} > 1,$$
$$k_{tr_{Ret}} + k_{tr_{Plt}} - k_{prol_{MEP}} > 0 \Rightarrow \frac{k_{tr_{Ret}} + k_{tr_{Plt}}}{k_{prol_{MEP}}} > 1,$$
$$k_{tr_{CMP}} + k_{tr_{MEP}} - k_{prol_{MPP}} > 0 \Rightarrow \frac{k_{tr_{CMP}} + k_{tr_{MEP}}}{k_{prol_{MPP}}} > 1.$$

Equation S6

Imposing conditions of Equation S5 and Equation S6 during parameter estimation required the use of constrained optimization algorithms with numerical boundaries. Therefore, we applied the following reparameterization:

$$\lambda_1 = \frac{T2_{Ret_0}}{T1_{Ret_0}} = \frac{Ret_0 k_{circ_{Ret}}}{T1_{Ret_0}\,a_{Ret}} > 1$$
$$\lambda_2 = \frac{T2_{Plt_0}}{T1_{Plt_0}} = \frac{Plt_0 k_{circ_{Plt}}}{T1_{Plt_0}\,a_{Plt}} > 1$$
$$\lambda_3 = \frac{k_{tr_{Neut}} + k_{tr_{Mono}}}{k_{prol_{CMP}}} = \frac{1}{k_{prol_{CMP}}}\left(k_{circ_{Neut}}\frac{Neut_0}{CMP_0} + k_{circ_{Mono}}\frac{Mono_0}{CMP_0}\right) > 1$$
$$\lambda_4 = \frac{k_{tr_{Ret}} + k_{tr_{Plt}}}{k_{prol_{MEP}}} = \frac{1}{k_{prol_{MEP}}}\left(\frac{a_{Ret}^2 T1^2{}_{Ret_0}}{k_{circ_{Ret}} Ret_0 MEP_0} + \frac{a_{Plt}^2 T1^2{}_{Plt_0}}{k_{circ_{Plt}} Plt_0 MEP_0}\right) > 1$$
$$\lambda_5 = \frac{k_{tr_{CMP}} + k_{tr_{MEP}}}{k_{prol_{MPP}}} = \frac{k_{tr_{CMP}} + k_{tr_{MEP}}}{k_{prol_{MPP}}} > 1$$

Equation S7

Once these new parameters (λ_i, $i=1,\ldots,5$) were estimated, we could then derive $T1_{Ret_0}$, $T1_{Plt_0}$, $k_{tr_{CMP}}$, $k_{tr_{MEP}}$, and k_{stem} from Equation S3 and Equation S4.

S8. Modelling and simulation approach

Simulations were performed in Matlab (2017b) toolbox Simbiology using a routine to deal with stiffness (function *ode15s*).

PK and PD models were fitted sequentially to rat data. We fitted first the PK model to rat free platinum PK data using Phoenix NLME 7.0 (Certara) with naïve pooled method, and multiplicative residual errors. Then, we fitted the model described in *Equation 1 – 9* (main text) to the rat bone marrow progenitor and peripheral blood counts pooled together from all studies (Table S1) using Matlab 17b (Mathworks) toolbox Simbiology (naïve pooled approach), with non-linear least square optimization (function *lsqnonlin)* and log-additive residual errors to correct data heteroscedasticity. Estimated parameter values are reported in Table 1, Table 2 of main text, and Table S3, Table S4.

We used Oscill8 Dynamical Systems Toolset (Copyright (C) 2005, Emery Conrad) to perform stability analysis on our model. We found that the model has two Hopf bifurcation points, induced by the parameter $\gamma_{prol_{Trans}}$ (4.6…, and 7.6…), Figure S7. For platelets, the homeostatic equilibrium becomes a limit cycle and then it loses stability after the first critical point (~4.6). For reticulocytes, oscillations occur after the second critical point (~7.6). These stability thresholds (i.e. $\gamma_{prol_{Trans}}$ values) depend on both platelet maturation time (MTT_{Plt}) and on the ratio between the size of $T2_{Plt}$ and $T1_{Plt}$ compartments (Figure S7).

S9. Scaling physiology-dependent system parameters

Physiology-dependent system parameters describe the biological properties of the system, which are independent of the drug. Examples are maturation times, population compartment sizes, cell circulating rates, etc. When scaling our rat model to human, all these values were updated taking into account cross-species difference.

We found baseline values for peripheral blood counts and mature cell circulating rates in the literature, while we made assumptions for the remaining parameters. Specifically, (i) feedback powers (γ) and new parameters λ were fixed to values estimated from rat data, (ii) baseline values for bone marrow progenitor cells were scaled allometrically from rat to human, and (iii) maturation times were derived from values estimated with a previously reported model [2].

A. Baseline values of bone marrow progenitor cells

Given the invasiveness of bone marrow biopsy in patients, data on bone marrow progenitors was not abundant in the literature. Moreover, cell surface markers used in flow cytometry analyses can be very specific, making the comparison rat vs human non- trivial. Therefore, we applied allometry to scale the size of MPP population from rat to human [3], using

$$MPP = \alpha M^{3/4},$$

Equation S8

where α is the allometric constant (*Log10(α) = 7.46*) and M is the average human body mass (70 Kg).

Other baseline progenitor values (i.e. CMP_0 and MEP_0) were derived from the allometrically scaled MPP size (MPP_0) assuming the following proportions between progenitor populations [4]:

$$CMP_0 / MPP_0 = 16,$$
$$MEP_0 / MPP_0 = 11.$$

Equation S9

B. Mean transit times

Mean transit times (*MTT*) can be found in literature from analyses performed on clinical data using Friberg model [2, 5], where MTT was defined as

$$MTT = {(n+1)}/{k_{circ}}$$

Equation S10

and n is the number of transit compartments included in the model, and k_{circ} is the circulating rate of mature cells. As discussed in [6], we assumed that maturation times are independent of cell production rates, and we defined $MTT = n/a$, where a is maturation rate within transit compartments, (Equation 6, Equation 8 main text). Therefore, before incorporating in our model human *MTT* values estimated with Friberg approaches [2, 5], we must apply the following correction factor [6] to these values

$$MTT = \left({n}/{n+1} \right) MTT_{Friberg} ,$$

Equation S11

which accounts for the de-coupling of progenitor and maturing compartments [6].

S10. Scaling drug-dependent system parameters

As described in Friberg *et al*. [7], drug effects can be scaled from rat to human taking into account species-specific drug sensitivity, i.e. applying the scaling reported in *Equation 10* of the main text. The potency values $IC50_R$ and $IC50_H$ were generated via colony-forming unit assays (Table S2).

Colony-forming unit assays

CD34+ cells were supplied by Poietics (human) or Harlan UK (RccHan:WIST rat). Cells were counted (Countess, Invitrogen) and diluted to a 10x concentration of 7.5 x 10^3/mL with IMDM containing 2% FCS. A final concentration of 750 cells/dish was required. A 4ml aliquot of Methocult (StemCell) was spiked with carboplatin (final concentration range 123 nmoles/L to 10 µmoles/L) and then 0.4 mL aliquot of cells added. 1.1 mL of cells/MethoCult was dispensed to each 35mm dish. Plates were incubated in a humidified chamber at 37ºC 95% air/5% CO_2 for 12-14 days and colonies (myeloid or erythroid) were counted using a light microscope.

S11. Carboplatin pharmacokinetics in human plasma

Human carboplatin pharmacokinetic was described with the following 2-compartment PK model previously published by Zandvliet *et al*. in [8].

We simulated a single dose of carboplatin, based on renal function and a targeted serum AUC, calculated accordingly to Calvert formula [9]:

$$Dose(mg) = AUC \times (GFR + 25) ,$$

Equation S12

where we assumed a target $AUC_{(0\text{-}24h)}$ of 5 (mg.min)/mL, and a glomerular filtration rate (GFR) of 125 mL/min (i.e. normal renal function).

S12. Description of Friberg model [2] used in [5] to analyse carboplatin-induced neutropenia and thrombocytopenia

To check the goodness of our clinical predictions, we compared our model results with simulated clinical data generated with Friberg model and parameter values reported in [5].

No PK data analysis, or individual creatinine clearance data and doses, were reported in [5], so we simulated the same PK per each patient, following a single dose of carboplatin corresponding to an $AUC_{(0\text{-}24)}$ of 5 [9]; see supplementary text S11.